Huawei Access Node KR-ACC-15 Optical Transport Failure SOP (ALARM_2205)

Node: KR-ACC-15 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_2205

Description

This Standard Operating Procedure (SOP) addresses the Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) in the Access Node KR-ACC-15, which is a Huawei NE40E platform. The Optical Transport Failure is a critical failure domain in this specific role, as it can impact the entire network's reliability and service availability. The failure is primarily attributed to the degradation of the SFP/XFP modules, fiber cuts, or reduced Rx/Tx power levels. In the Huawei NE40E platform, the optical transport system is responsible for transmitting and receiving optical signals between the access node and other network devices. The SFP/XFP modules are used to convert electrical signals to optical signals, and their degradation or failure can lead to signal loss, distortion, or even complete loss of connectivity. The fiber cuts or reduced Rx/Tx power levels can also cause signal attenuation, leading to optical transport failure. The NE40E's optical transport system operates based on the Optical Transport Network (OTN) protocol, which provides a framework for managing and monitoring the optical transport network. The OTN protocol ensures that the optical signals are properly demultiplexed, multiplexed, and transported between the access node and other network devices. However, the degradation of the SFP/XFP modules or fiber cuts can compromise the OTN protocol's functionality, leading to optical transport failure.

Pre-requisite Checks

- Verify that the KR Puram Access site is accessible and the physical infrastructure is intact.

- Check the redundant link status to ensure that the network has an alternative path for traffic flow.

- Verify that the Huawei NE40E platform's power supply is stable and within the recommended operating range.

- Check the environment for any signs of physical damage, water ingress, or extreme temperature fluctuations.

Potential Root Causes

Cause	Probability
Degraded SFP/XFP modules	High
Fiber cuts	Medium
Reduced Rx/Tx power levels	Low

Diagnostic Commands & Expected Outputs

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Command: display optical-transport interface all, Expected Output: Interface information, including the SFP/XFP module status, Rx/Tx power levels, and optical signal quality., Description: This command displays the interface information for all optical interfaces in the system, including the SFP/XFP module status, Rx/Tx power levels, and optical signal quality. This provides a comprehensive view of the optical transport system's status and helps identify potential issues.

Command: display optical-transport alarm, Expected Output: Alarm information, including the alarm type, severity, and affected interface., Description: This command displays the alarm information for the optical transport system, including the alarm type, severity, and affected interface. This helps identify the specific issue causing the optical transport failure and isolate the affected component.
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Resolution Steps

Step 1: Step: 1. Replace the degraded SFP/XFP module, Command: replace optical-transport interface with , Description: Replace the degraded SFP/XFP module with a new one to restore the optical transport system's functionality. Ensure that the new module is properly seated and securely fastened to the interface.

Step 2: Step: 2. Repair or replace the fiber cable, Command: repair or replace fiber cable , Description: Repair or replace the fiber cable to restore the optical transport system's functionality. Ensure that the fiber cable is properly routed and secured to prevent further damage.

Step 3: Step: 3. Adjust the Rx/Tx power levels, Command: adjust optical-transport interface power , Description: Adjust the Rx/Tx power levels to optimize the optical transport system's performance. Ensure that the new power level is within the recommended operating range to prevent damage to the SFP/XFP module or fiber cable.

Post-Check Verification

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Command: display optical-transport interface all, Expected Output: Interface information, including the SFP/XFP module status, Rx/Tx power levels, and optical signal quality., Description: Verify that the optical transport system is functioning properly by checking the interface information, including the SFP/XFP module status, Rx/Tx power levels, and optical signal quality.

Command: display optical-transport alarm, Expected Output: Alarm information, including the alarm type, severity, and affected interface., Description: Verify that the optical transport system is free from alarms and that the issue causing the optical transport failure has been resolved.
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Roll-back Procedures

- Step: 1. Restore the original SFP/XFP module, Command: replace optical-transport interface with , Description: Restore the original SFP/XFP module to its previous state to ensure that the optical transport system's functionality is not compromised.

- Step: 2. Revert the fiber cable repair or replacement, Command: revert fiber cable , Description: Revert the fiber cable repair or replacement to its previous state to ensure that the optical transport system's functionality is not compromised.

- Step: 3. Revert the Rx/Tx power level adjustment, Command: adjust optical-transport interface power , Description: Revert the Rx/Tx power level adjustment to its previous state to ensure that the optical transport system's functionality is not compromised.